

Data Dictionary

Companion to *TECHNICAL_HANDOVER.md* Part VII

Snapshot date: 2026-05-10 **Schema source:** `bills.db` (live production) **Generation:** `PRAGMA table_info(<table>)` per table.

How to read this

Each table below lists every column with its SQL type and any constraints (`PK` = primary key, `NOT NULL`, `default=`). Type `TEXT` is the SQLite default and is used for dates (ISO 8601 strings), enums, and most string fields. `INTEGER` columns with default 0 or 1 are boolean flags in practice. `REAL` is used for money and decimal quantities.

users

Col	Type	Notes
<code>username</code>	TEXT	PK
<code>password_hash</code>	TEXT	NOT NULL — PBKDF2-HMAC-SHA256, 200k iterations, base64(salt+hash)
<code>full_name</code>	TEXT	
<code>role</code>	TEXT	default= <code>operator</code> — also: <code>admin</code>
<code>is_active</code>	INTEGER	default=1 — soft-disable users without losing audit linkage
<code>must_change_password</code>	INTEGER	default=1 — forces /change-password redirect on first login
<code>created_at</code>	TEXT	
<code>last_login</code>	TEXT	

bills

Col	Type	Notes
id	INTEGER	PK
bill_no	TEXT	NOT NULL — human-readable, e.g. JL-0008
bill_date	TEXT	
recipient_name	TEXT	The consignor / paying party
recipient_address	TEXT	
recipient_gstin	TEXT	
state_code	TEXT	2-digit GST state code (e.g. 09 = UP)
trip_type	TEXT	One Way / Two Way
vehicle_no	TEXT	
freight_type	TEXT	e.g. A1/A4 — consignor's internal freight code
delivery_month	TEXT	MMM/YY, displayed in bill footer
client_name	TEXT	Display-only override for tax-footer text
total_amount	REAL	default=0 — computed sum of deliveries
deliveries	TEXT	default= [] — JSON array of delivery line items
created_at	TEXT	
ledger_entry_id	INTEGER	FK → ledger_entries.id (for single-LR bills)
client_paid	INTEGER	default=0
client_paid_date	TEXT	
client_paid_amount	REAL	
client_paid_mode	TEXT	Cash / Bank / UPI / Cheque
client_paid_reference	TEXT	

deliveries JSON shape (per line item):

```
{
  "gr_no": "12571",
  "outward_no": "880207639",
  "outward_date": "2026-05-03",
  "inward_no": "",
  "inward_date": "",
  "location": "DIBIYAPUR",
  "consignee": "VAIBHAV LAXMI ENTERPRISES",
  "delivery_qty": 1100,
  "converted_case": 1100,
  "inward_qty": "",
  "empty_qty": "",
  "weight": 20110.5,
  "freight_rate": 6233.82,
  "overload": 0,
  "toll_tax": 0,
  "excess_km": 0,
  "detention": 0,
  "unloading": 0,
  "value_of_supply": 6234
}
```

challans

Col	Type	Notes
id	INTEGER	PK
lr_no	TEXT	The LR / GR number — printed pre-numbered on the challan slip
challan_date	TEXT	
consignor_name	TEXT	Who hires JL — typically Shri Shiv Shakti
consignor_address	TEXT	
consignee_name	TEXT	Destination party
consignee_address	TEXT	
from_city_state	TEXT	
to_city_state	TEXT	
invoice_no	TEXT	Consignor's invoice number, appears on PoD middle slash
invoice_date	TEXT	
consignment_value	REAL	
gst_number	TEXT	
no_of_articles	TEXT	Free text — can be "610 nos" or "1 lot"
description	TEXT	
value_of_goods	REAL	
weight_kg	REAL	
del_no	TEXT	JL's outward delivery number — usually = consignor PoD Doc No.
shipment_no	TEXT	Optional alt identifier
cost_no	TEXT	Optional alt identifier
seal_no	TEXT	Tamper seal applied at dispatch
driver_name	TEXT	
driver_mobile	TEXT	
truck_no	TEXT	Normalised: uppercase, no spaces
gate_in_time	TEXT	
gate_out_time	TEXT	
lane_transit_time	TEXT	
expected_arrival	TEXT	
source_image	TEXT	Relative path under uploads/, the photo we extracted
raw_extraction	TEXT	Full Gemini JSON response (for forensics)

Col	Type	Notes
confidence_json	TEXT	{field: 'low'\ 'medium'} for fields the AI flagged
status	TEXT	default= open — also: draft , pod_received , billed
notes	TEXT	
created_at	TEXT	
updated_at	TEXT	
ledger_entry_id	INTEGER	FK → ledger_entries.id (cross-stage Phase D link)
pod_doc_no	TEXT	What the buyer's PoD slip Doc No. says — should match del_no

ledger_entries

Col	Type	Notes
id	INTEGER	PK
challan_id	INTEGER	FK → challans.id (nullable for manually-entered)
entry_date	TEXT	
gr_no	TEXT	= challans.lr_no in practice
vehicle_no	TEXT	
station	TEXT	Destination
shipment_no	TEXT	Often = challans.invoice_no
trip_type	TEXT	default= One Way
mt_qty	REAL	Tonnage (1 mt = 1000 kg)
freight	REAL	default=0 — what JL earns for this trip
advance_cash	REAL	default=0 — cash given to driver
advance_account	REAL	default=0 — account-transfer advance
diesel	REAL	default=0 — diesel cost (charged to truck)
diesel_vendor_id	INTEGER	FK → diesel_vendors.id
transporter_id	INTEGER	FK → transporters.id (null = family-owned truck)
pod_received	INTEGER	default=0
pod_date	TEXT	
pod_image	TEXT	Path under uploads/pods/
paid	INTEGER	default=0 — driver/transporter settled?
paid_date	TEXT	
paid_mode	TEXT	
paid_amount	REAL	
paid_reference	TEXT	
remarks	TEXT	
created_at	TEXT	
updated_at	TEXT	
bill_id	INTEGER	FK → bills.id (Phase D link)
weight_kg	REAL	Optional ledger-side weight, for cross-verification with challan

Computed: `balance = freight - advance_cash - advance_account - diesel` (computed in Python, not stored)

payments

Col	Type	Notes
id	INTEGER	PK
party_type	TEXT	NOT NULL — client / transporter / diesel_vendor
party_key	TEXT	NOT NULL — recipient name for client, id for others
payment_date	TEXT	NOT NULL
amount	REAL	NOT NULL — always positive; direction implied by party_type
mode	TEXT	Cash / Bank / UPI / Cheque
reference	TEXT	
notes	TEXT	
source	TEXT	default= manual — also: migrated
created_at	TEXT	
created_by	TEXT	

audit_log

Col	Type	Notes
id	INTEGER	PK
occurred_at	TEXT	NOT NULL
user_name	TEXT	From current_user() session
action	TEXT	create / update / delete / pod_mark / bulk_pod / reset_password / ...
entity	TEXT	bill / challan / ledger_entry / payment / user
entity_id	INTEGER	
summary	TEXT	Human-readable one-liner
changes	TEXT	JSON: {field: [old, new], ...}

Indexes: idx_audit_entity on (entity, entity_id); idx_audit_when on occurred_at DESC.

transporters

Col	Type	Notes
id	INTEGER	PK
name	TEXT	NOT NULL, UNIQUE
mobile	TEXT	
bank_details	TEXT	
notes	TEXT	
created_at	TEXT	
updated_at	TEXT	

diesel_vendors

Col	Type	Notes
id	INTEGER	PK
name	TEXT	NOT NULL
location	TEXT	
notes	TEXT	
created_at	TEXT	
updated_at	TEXT	

recipients

Col	Type	Notes
name	TEXT	PK
address	TEXT	
gsttin	TEXT	
state_code	TEXT	
updated_at	TEXT	
freight_rate	REAL	Last-billed rate cached for fallback when rate list misses

vehicles

Col	Type	Notes
vehicle_no	TEXT	PK — normalised: uppercase, no spaces
updated_at	TEXT	
transporter_id	INTEGER	FK → transporters.id (which owner does this truck belong to)

drivers

Col	Type	Notes
mobile	TEXT	PK
name	TEXT	
updated_at	TEXT	

freight_rates (the rate list)

Col	Type	Notes
id	INTEGER	PK
customer_name	TEXT	The consignee (destination party) — NOT the bill recipient
party_code	TEXT	
location	TEXT	Destination town/city
dist_twy_km	INTEGER	Two-way distance
dist_owy_km	INTEGER	One-way distance
lp_owy	REAL	LP truck, one-way rate
lp_twy	REAL	LP truck, two-way rate
trolla_owy	REAL	Trolla truck, one-way rate
trolla_twy	REAL	Trolla truck, two-way rate
updated_at	TEXT	

Uniqueness: `UNIQUE(customer_name, location)` — one consignee can have multiple destinations.

extractions

Col	Type	Notes
id	INTEGER	PK
created_at	TEXT	
mode	TEXT	combine / split — how to assemble multi-photo extractions
status	TEXT	pending / extracted / reviewed / used
note	TEXT	

extracted_invoices

Col	Type	Notes
id	INTEGER	PK
extraction_id	INTEGER	FK → extractions.id
file_name	TEXT	Path under uploads/
seq	INTEGER	Order in batch
raw_json	TEXT	Gemini's raw output
edited_json	TEXT	After user review
error	TEXT	

ledger_extractions

Col	Type	Notes
id	INTEGER	PK
source_image	TEXT	
page_date	TEXT	The date written on the ledger page header
raw_json	TEXT	
edited_json	TEXT	
status	TEXT	default= pending
created_at	TEXT	

settings (singleton-ish)

Col	Type	Notes
key	TEXT	PK
value	TEXT	

Keys currently in use:

- `next_bill_number` — auto-incrementing counter for bill numbering
- `next_lr_number` — auto-incrementing LR counter
- `pod_overdue_days` — default 10
- `default_consignor_name`, `default_consignor_address`, `default_consignor_gstin`, `default_consignor_state` — pre-fill new bills with the family's main consignor
- `client_name` — display name for the tax footer
- `vehicle_type` — `LP` / `Trolla` default
- `freight_type` — `A1/A4` default
- `clients` — JSON array of allowed client names (for the picker)
- `recipient_name_consignor_fix` — migration flag, ignore in queries

Indexes summary

```
CREATE INDEX idx_audit_entity      ON audit_log(entity, entity_id);
CREATE INDEX idx_audit_when       ON audit_log(occurred_at DESC);
CREATE INDEX idx_bills_date       ON bills(bill_date DESC);
CREATE INDEX idx_bills_recipient  ON bills(recipient_name);
CREATE INDEX idx_bills_paid       ON bills(client_paid);
CREATE INDEX idx_ledger_date      ON ledger_entries(entry_date DESC);
CREATE INDEX idx_ledger_pod       ON ledger_entries(pod_received, bill_id);
CREATE INDEX idx_ledger_vehicle   ON ledger_entries(vehicle_no);
CREATE INDEX idx_ledger_transp    ON ledger_entries(transporter_id);
CREATE INDEX idx_challans_lr      ON challans(lr_no);
CREATE INDEX idx_challans_status  ON challans(status);
CREATE INDEX idx_rates_customer  ON freight_rates(customer_name);
```

These were chosen based on observed query patterns. The CTO should rerun `EXPLAIN QUERY PLAN` after every major schema change to confirm new queries hit indexes.